

CSCE 645 GEOMETRIC MODELING · MODULE 06 · LECTURE NOTES

Subdivision Surfaces

Smoothness by refinement, with no patches to join.

MODULE OUTCOMES

- Implement Catmull–Clark and Loop subdivision.
- Explain the relationship between subdivision and spline surfaces.
- Explain extraordinary vertices and what happens there.
- Implement creases and semi-sharp features.
- Explain adaptive and feature-adaptive subdivision.

1 Refinement as a definition

Module 05 ended at an impasse: tensor-product patches cannot cover arbitrary topology smoothly, and the vertex enclosure problem means the difficulty is structural rather than a matter of effort.

Subdivision takes a different approach. Rather than defining the surface by a formula, define it as the **limit** of repeatedly refining a coarse control mesh by a fixed local rule. There are no patches, hence no patch boundaries, hence no continuity conditions to impose — the smoothness is a property of the limit process itself.

Scheme	Base mesh	Limit surface	Notes
Catmull–Clark	Quads; handles any polygons	C^2 everywhere except at extraordinary vertices, where C^1	The dominant scheme. Film and games standard.
Loop	Triangles	C^2 except C^1 at extraordinary vertices	Natural for triangle-based pipelines.
Doo–Sabin	Any polygons	C^1	Historically important, rarely used now.
$\sqrt{3}$	Triangles	C^2	Slower refinement ratio, so finer control over level of detail.

1.1 Catmull–Clark

FACE POINT	for each face: average of its vertices.
EDGE POINT	for each edge: average of its two endpoints and the two adjacent face points.
VERTEX POINT	for each original vertex of valence n , move it to $(F + 2R + (n-3)P) / n$ where F = average of adjacent face points, R = average of adjacent edge midpoints, P = the original position.
CONNECT	join each face point to the edge points of its face.

Two structural facts fall out immediately. First, **after one step every face is a quadrilateral**, regardless of the input — a pentagon becomes five quads. Second, every newly created vertex has valence 4 (new face points) or valence 4 (new edge points on a quad mesh), so **all irregularity remains at the original vertices**.

2 Why it works

CATMULL-CLARK ON A REGULAR QUAD MESH IS A BICUBIC B-SPLINE

Take a mesh in which every vertex has valence 4. Apply Catmull–Clark. The limit surface is *exactly* the uniform bicubic B-spline surface over that control grid — the object from Module 05.

This is not an approximation or an analogy. The subdivision rules were derived from the B-spline refinement relations, which is why they look arbitrary until you know where they came from.

So subdivision is not a competitor to spline surfaces. It is their **extension to arbitrary topology**: a procedure that agrees exactly with the spline wherever a spline could be defined, and continues to produce a sensible smooth surface where one could not.

The special behaviour at extraordinary vertices is precisely the price of that extension, and it is a small price.

2.1 Extraordinary vertices

A vertex is **extraordinary** (or irregular, or a pole) if its valence differs from the regular value: 4 for quad schemes, 6 for triangle schemes. Module 01's Euler argument showed these cannot always be avoided — a closed surface of genus 0 cannot be made entirely of valence-4 quads.

- Their number and location are fixed by the base mesh. Subdivision never creates new extraordinary vertices.
- After one subdivision step, every extraordinary vertex is surrounded by regular vertices, so the irregularity becomes **isolated** — which is what makes feature-adaptive evaluation (§4) possible.
- At an extraordinary vertex the limit surface is C^1 but not C^2 . Position and tangent plane are continuous; curvature is not.

WHAT A C^1 POINT LOOKS LIKE

A curvature discontinuity is invisible in silhouette and clearly visible in a reflection — exactly as Module 03 described for curves.

On a car body or any polished surface, an extraordinary vertex in the middle of a smooth panel shows as a subtle distortion in the reflected environment. This is why modelling guidelines insist on keeping quad topology regular across large smooth areas and placing poles where they will not be seen — inside an ear, under an arm, at a hard material boundary.

It is also why zebra-stripe analysis exists in modelling packages: it makes curvature discontinuities visible before rendering.

3 Creases

Subdivision smooths everything by default, and real objects have sharp features. Two mechanisms address this.

3.1 Sharp creases

Tag an edge as a crease. Along tagged edges, apply *curve* subdivision rules — treating the crease as a B-spline curve — rather than surface rules. The limit surface then has a genuine tangent discontinuity there. Tagging a vertex similarly produces a corner.

3.2 Semi-sharp creases

A binary sharp/smooth choice is too coarse. **Semi-sharp** creases assign a sharpness value s to an edge: apply crease rules for the first s subdivision levels, then revert to smooth rules.

WHY THIS WAS THE FEATURE THAT MATTERED

A perfectly sharp edge reads as computer-generated, because real manufactured edges have a small fillet from tooling, moulding, or wear. The eye notices its absence.

Modelling that fillet explicitly requires extra edge loops along every hard edge, which bloats the control cage and makes it painful to edit.

A semi-sharp crease produces the same appearance from a single number per edge, with no additional geometry. Fractional sharpness values interpolate between levels, so an artist can dial an edge continuously from soft to knife-sharp.

Pixar introduced this (DeRose et al., 1998) and it is the development that made subdivision practical for hard-surface modelling rather than only organic forms. Every serious implementation has it.

4 Evaluating a subdivision surface

Method	How	Trade-off
Iterative refinement	Apply the rules k times and render the resulting mesh.	Simple and obviously correct. Face count grows as 4^k , so four levels is $256\times$. Memory and time become prohibitive quickly.
Exact evaluation (Stam)	Use the eigenstructure of the subdivision matrix to evaluate the limit surface directly at any (u,v) , including in the neighbourhood of an extraordinary vertex.	Exact and arbitrary-resolution. Mathematically involved, and the eigen-decomposition must be precomputed per valence.
Feature-adaptive	Refine only near extraordinary vertices and creases. Everywhere else the mesh is regular, so the surface is already a bicubic B-spline patch and can be evaluated directly.	The practical answer. This is what OpenSubdiv does, and it is why film-quality subdivision runs at interactive rates.

FEATURE-ADAPTIVE SUBDIVISION IS THE PAYOFF OF §2

Because regular regions are *exactly* bicubic B-spline patches, they need no refinement at all — they can be handed to the GPU tessellator and evaluated as patches at whatever resolution the view requires (CSCE 641 Module 12).

Only the small neighbourhoods around extraordinary vertices and creases need explicit refinement, and because irregularity is isolated after one step, those neighbourhoods are small and do not grow.

So the cost is dominated by the number of poles in the cage, not by the subdivision level. A character with a few dozen extraordinary vertices evaluates in real time at film resolution. This is the single most important practical fact in the module.

5 Subdivision against the alternatives

	NURBS patches	Subdivision	Plain triangle mesh
Arbitrary topology	Difficult — needs patch layout and trimming	Free	Free
Smoothness	Exact, analytic	C^2, C^1 at extraordinary vertices	None
Local editing	Affects a whole patch	Local	Local
Level of detail	Re-tessellate	Intrinsic — just subdivide less	Author an LOD chain (Module 10)
Exact conics	Yes	No	No
Authoring effort	High — patch layout is skilled work	Low — model a quad cage	Medium

Subdivision wins on every row except exactness, and exactness is precisely what CAD cannot give up. That single column is the whole explanation for why the two representations coexist rather than one displacing the other.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Keenan Crane — DDG, subdivision material	The rules derived rather than stated, with the B-spline connection made explicit.
DeRose, Kass & Truong — Subdivision Surfaces in Character Animation (free)	The Pixar paper that introduced semi-sharp creases and put subdivision into production. Short and very readable.

Resource	What it covers for this module
Jos Stam — Exact Evaluation of Catmull-Clark Subdivision Surfaces (free)	How to evaluate the limit surface directly. The eigenstructure argument is worth working through once.
OpenSubdiv (Pixar, open source)	The production implementation, with excellent documentation of feature-adaptive subdivision. Read the overview even if you do not use the library.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement Catmull–Clark subdivision on your half-edge structure. Apply it to a cube three times and verify visually that it converges toward a sphere-like shape.
- 2 Verify the B-spline equivalence: build a regular 4×4 quad grid, subdivide it several times, and compare the result against a bicubic B-spline surface evaluated over the same control points. Report the maximum deviation.
- 3 Identify and visualise the extraordinary vertices of a subdivided cube (there are eight, of valence 3). Confirm that their number does not change with subdivision level.
- 4 Render a subdivided surface with a mirror material and an environment map. Find the reflection distortion at an extraordinary vertex and capture it.
- 5 Implement Loop subdivision for triangle meshes and compare the limit surfaces of Loop and Catmull–Clark on the same shape.
- 6 Implement sharp creases, then semi-sharp creases with an integer sharpness. Render a cube with sharpness 0, 2, 4, and infinite, and compare.
- 7 Implement feature-adaptive subdivision: detect regular regions and evaluate them as B-spline patches, refining only near poles. Measure the reduction in generated geometry against uniform subdivision at the same quality.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does subdivision avoid the vertex enclosure problem entirely?
- 2 State the three Catmull–Clark rules and two structural facts that follow after one step.
- 3 What is the relationship between Catmull–Clark and bicubic B-spline surfaces, and why does it matter?
- 4 What is an extraordinary vertex, what is the surface's continuity there, and how is it visible?
- 5 Why are semi-sharp creases more useful than binary sharp edges?

- 6 Explain feature-adaptive subdivision and why it depends on the fact in question 3.
- 7 Which single column does subdivision lose to NURBS on, and why does that keep both alive?